Chapter 9, Problem 2RQ

Problem

A function that repeats itself after fixed intervals is said to be:

(a) a phasor

(b) harmonic

(c) periodic

(d) reactive

Step-by-step solution

Step 1 of 4

A complex number that defines amplitude and phase of a sinusoid is known as phasor.

Therefore, option (a) is incorrect.

Step 2 of 4

If the frequency of a signal or wave is an integral multiple of the frequency of another signal or a wave, then such signal is known as harmonic.

Therefore, option (b) is incorrect.

Step 3 of 4

A function is said to be reactive, if it is purely imaginary.

Therefore, option (d) is incorrect.

Step 4 of 4

A function $f(t)$ is said to be periodic, if it satisfies the following condition:

$$f(t) = f(t + nT)$$

Here, *n* is any integer and *T* is the time period.

Hence, the function $f(t)$ repeats itself after fixed intervals.

Therefore, the correct option is **(c).**